VELOOP REWARDS - Data Quality Audit

Data Analyst Internship, Task 01 | Deliverable 13 | Work Package K

SIMULATED DATA - CREATED FOR ANALYTICAL TESTING. The audited file is fictional; every count describes the supplied simulated dataset only and is not an actual VELOOP statistic. Personal-looking fields are not reproduced.

1. Summary

- **25 issues logged**: 0 Critical, 6 High, 15 Medium, 4 Low. 12 come from the problems injected into the audit sheet and 13 are present in both sheets.
- **The audit sheet is the clean sheet plus 8 repeated rows and 12 changed cells** (8 blanks, 1 missing and 1 malformed registration time, and 2 registration times altered so they contradict the stored days_since_registration). All were found by the rules and traced to exact rows.
- **The 'clean' sheet is not rule-free.** It has negative derived days (4), logins after the period end (4), 72 users with sessions but no login, 30 users with XP but no session, 389 users with more active days than days available, and 34 marketing opt-ins while consent is Pending.
- **Rule coverage:** 18 of the 39 quality rules could be tested on the supplied file (7 passed, 8 failed, the rest partly or information); 21 could not be tested because the other datasets were not supplied.
- **Some problems are invisible to format checks.** Two registration times look valid but contradict the stored derived value; only a cross-field check (proposed rule DQ040) finds them.

2. The brief's 14 issue types

Issue type	Records found	Result	Where
duplicate IDs	8	Found	DQA-01
missing IDs	0	Not found	-
missing timestamps	1	Found	registration_datetime
malformed dates	1	Found (plus missing timezone on all values)	registration_datetime; timezone
invalid statuses	0	Not found (all values are in the controlled lists)	-
inconsistent capitalization	0	Not found	-
trailing spaces	0	Not found	-
invalid numeric values	0	Not found (other than blanks)	-
negative values where inappropriate	4	Found (derived field in both sheets)	days_since_registration
unexpected nulls	8	Found	8 fields
orphan records	0	Not found (messy to clean); other tables not supplied	-
duplicate events	0	Not testable: no event data supplied	-
inconsistent naming	0	Not found in values; see status and activity meaning issue	DQA status issue
impossible state transitions	616	Found as impossible combinations; history is not supplied, so true transitions cannot be tested	DQA state issues

Impossible state transitions are counted as record-issue pairs; one record can appear under several issues. True transitions cannot be tested because no status history was supplied.

3. Data quality report

ID	Dataset	Field	Issue type	Records	Severity	Detection logic	Likely cause	Correction	Prevention	Owner
DQA-01	Users_Messy_Audit	user_id (whole row)	duplicate IDs	8	High	COUNT(*) per user_id greater than 1; the repeated rows are exact copies of the first occurrence	Controlled test injection: the 8 repeated rows are appended at the end of the sheet (README: the audit sheet contains duplicates)	Keep the first occurrence of each user_id; log the repeats and hold them in quarantine	Unique constraint on user_id (DQ002); idempotent writes (DQ018)	Engineering
DQA-02	Users_Messy_Audit	city	unexpected nulls	1	Medium	city IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-03	Users_Messy_Audit	acquisition_source	unexpected nulls	1	Medium	acquisition_source IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-04	Users_Messy_Audit	primary_device	unexpected nulls	1	Medium	primary_device IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-05	Users_Messy_Audit	onboarding_status	unexpected nulls	1	Medium	onboarding_status IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-06	Users_Messy_Audit	email_verification_status	unexpected nulls	1	Medium	email_verification_status IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-07	Users_Messy_Audit	age	unexpected nulls	1	Medium	age IS NULL; dictionary marks it Required	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-08	Users_Messy_Audit	gender	unexpected nulls	1	Low	gender IS NULL; dictionary marks it Optional	Controlled test injection: one value blanked (the same value is present in the clean sheet)	Restore from the source system; if unavailable, set a documented 'unknown' value and flag	Required-field check at the application and at collection (DQ001, DQ015)	Engineering
DQA-09	Users_Messy_Audit	last_login_datetime	unexpected nulls	1	Medium	last_login_datetime IS NULL for a user whose account_status is not Pending Verification (in the dataset, never-logged-in users are exactly the Pending Verification users)	Controlled test injection: one login time blanked	Restore from the source system	Conditional-field rule: login time required once an account has left Pending Verification (DQ015)	Engineering
DQA-10	Users_Messy_Audit	registration_datetime	missing timestamps	1	High	registration_datetime IS NULL (required field)	Controlled test injection	Restore from the source system; the registration time anchors every cohort	Server-generated registration time written with the account (DQ004)	Engineering
DQA-11	Users_Messy_Audit	registration_datetime	malformed dates	1	High	Value is present but does not parse as a datetime	Controlled test injection: the text '2026-14-02 12:00' has a month of 14, so it looks like day and month were swapped	Do not guess the intended date (it could be 14 February); restore from the source	Reject non-ISO 8601 values at collection (DQ005)	Engineering
DQA-12	Users_Messy_Audit	registration_datetime vs days_since_registration	inconsistent derived value	2	High	Recompute days_since_registration from registration_datetime and compare; a difference above 0.2 days is flagged	Controlled test injection: registration_datetime was changed to another valid date but the stored derived value was not updated	Restore the registration time; recompute derived values instead of trusting them	Do not store derived values; compute them in the analytics layer, or recompute and compare on load (DQ040)	Data Analyst
DQA-13	Both sheets	days_since_registration	negative values where inappropriate	4	Medium	days_since_registration less than 0; the dictionary says it must be 0 or more	The reference time used for the derived value appears to be 15 Sep 2026 00:00, but four users registered later on 15 Sep, so their value is negative	Use a reference time at or after the latest registration, or compute from the actual load time	Check derived values against their own validation rule on load	Data Analyst
DQA-14	Both sheets	last_login_datetime	timestamps after the dataset period	4	Medium	last_login_datetime later than the stated period end (15 Sep 2026); 9 are later than the reference time 15 Sep 2026 00:00	Login times were generated without an upper bound at the period end	Cap or correct; confirm what the reference time means	Reject timestamps later than the receive time plus tolerance (DQ007)	Engineering

ID	Dataset	Field	Issue type	Records	Severity	Detection logic	Likely cause	Correction	Prevention	Owner
DQA-15	Both sheets	lifetime_sessions vs last_login_datetime	impossible state	72	High	lifetime_sessions greater than 0 while last_login_datetime is empty	All 72 are Pending Verification accounts; sessions were generated for them although they have no login	Decide which field is authoritative; correct the other	Counter consistency check: sessions greater than 0 requires a login time (DQ041)	Engineering
DQA-16	Both sheets	lifetime_sessions vs last_login_datetime and xp_balance	impossible state	30	Medium	lifetime_sessions equals 0 while a login time exists and xp_balance is above 0	Login time and XP were generated independently of the session counter	Decide which field is authoritative	Counter consistency check (DQ041)	Engineering
DQA-17	Both sheets	active_days	impossible state	389	Medium	active_days greater than the number of calendar days from registration to last login, inclusive	Active days were generated independently of the login window	Recompute active_days from activity records	Counter consistency check (DQ041)	Data Analyst
DQA-18	Both sheets	active_days vs lifetime_sessions	impossible state	91	Medium	active_days equals 0 while lifetime_sessions is above 0	Counters generated independently	Recompute active_days	Counter consistency check (DQ041)	Data Analyst
DQA-19	Both sheets	consent_status vs marketing_opt_in	impossible state	34	High	marketing_opt_in = Yes while consent_status = Pending	Consent and opt-in were generated independently	Treat opt-in as not valid until consent is Accepted; confirm with Legal	Consent consistency rule: opt-in requires accepted consent (DQ042)	Product / Legal
DQA-20	Both sheets	account_status vs is_active_30d	inconsistent naming and meaning	120	Medium	Status 'Suspended' or 'Inactive' with is_active_30d = Yes (25 suspended and 95 inactive accounts)	account_status is stored independently of the derived activity flags, and the word 'Inactive' is used both as a status and as an activity concept	Separate account state (stored) from activity (derived); rename or define the status	Dictionary definitions that distinguish state from activity (see the lifecycle model in Work Package B)	Product
DQA-21	Both sheets	onboarding_status vs lifetime_sessions	impossible state	100	Low	onboarding_status = Not Started while the user has sessions and a login	Possible meaning: onboarding is optional; or the status was generated independently	Confirm the rule with Product (A019)	Define when onboarding status changes	Product
DQA-22	Both sheets	account_status vs email_verification_status	unknown rule	73	Low	Accounts that are Active, Inactive or Suspended while the email is still Pending (73); all 72 Pending Verification accounts do have a Pending email	Unknown: it may be allowed for an account to be active before verification	Confirm the rule (A047)	Document the allowed combinations of status and verification	Product
DQA-23	Both sheets	registration_datetime, last_login_datetime	malformed dates	1928	Medium	Datetime values carry no timezone or offset (1,928 values in the clean sheet)	Timestamps exported without a timezone	State the timezone in the dictionary and store in UTC	Require an explicit offset or UTC at collection (DQ006)	Engineering
DQA-24	Both sheets	platform, country	undocumented and constant fields	1000	Low	platform is 'Web' for every row and country is 'India' for every row; neither column is listed in the dataset's own Data_Dictionary (25 fields documented, 27 columns present)	Dataset design; the two columns were added without dictionary entries	Keep, but do not rely on them for segmentation; platform differs from primary_device (Android, iOS, Windows, macOS)	Record app platform separately from device	Product
DQA-25	Both sheets	full_name, email, age, gender, state, city	personal-looking fields	6	Medium	The dataset holds name, email, age, gender, state and city, which the proposed data dictionary deliberately does not collect	Dataset design (simulated)	Do not carry them into analytics; use the pseudonymous user_id; share only a sanitised version with interns	Data minimisation (Work Package P); approved or sanitised datasets for interns	Product

4. Rule coverage on the supplied data

Rule	Name	Primary layer	Result	Detail	Issue ID(s)	Note
DQ001	Identifier present	Application	Pass	0 missing user_id values	-	
DQ002	Primary key unique	Application	Fail	8 repeated user_ids (audit sheet)	DQA-01	
DQ003	Identifier format	Data Collection	Pass	0 values break the user_id or referral_code pattern	-	
DQ004	Timestamp present	Application	Fail	1 missing registration_datetime	DQA-10	
DQ005	Timestamp format	Data Collection	Fail	1 malformed registration_datetime	DQA-11	
DQ006	Timezone explicit	Data Collection	Fail	No datetime value carries a timezone	DQA-23	
DQ007	No future timestamps	Data Collection	Fail	4 logins after the stated period end (9 after the reference time)	DQA-14	
DQ008	Timestamp order	Transformation	Pass	0 users with a login earlier than the registration	-	
DQ009	Allowed status values	Application	Pass	0 values outside the controlled lists	-	The lists were taken from the clean sheet because no approved list exists yet (A022)
DQ010	Capitalization standard	Transformation	Pass	0 capitalization variants	-	
DQ011	No stray whitespace	Transformation	Pass	0 values with stray spaces	-	
DQ012	Numeric value valid	Application	Pass	0 non-numeric values (blanks aside)	-	
DQ013	Amounts are positive	Application	Fail	4 negative days_since_registration (rule applied to all non-negative numbers)	DQA-13	DQ013 is written for amounts; the audit extends it to non-negative numeric fields
DQ014	Unit present and approved	Application	Not testable	No unit fields in the supplied data	-	Needs reward, wallet or withdrawal data
DQ015	Conditional fields present	Data Collection	Fail	8 unexpected blanks in required or conditional fields	DQA-02, DQA-03, DQA-04, DQA-05, DQA-06, DQA-07, DQA-09	
DQ016	User exists	Transformation	Partly	0 orphans between the audit and clean sheets; no other tables to check	-	Only one relationship exists in the file
DQ017	Other foreign keys exist	Transformation	Not testable	No other tables supplied	-	Needs tasks, rewards, wallet, withdrawals and referrals
DQ018	Duplicate wallet transaction	Application	Not testable	No wallet data	-	
DQ019	Duplicate event	Data Collection	Not testable	No event data	-	Repeated users are found by DQ002
DQ020	Event name in catalogue	Data Collection	Not testable	No event data	-	
DQ021	Field names follow dictionary	Transformation	Fail	2 columns (platform, country) are not in the dataset's own dictionary; status and activity meaning overlap	DQA-20, DQA-22, DQA-24	
DQ022	Event properties complete	Data Collection	Not testable	No event data	-	
DQ023	Schema version known	Data Collection	Not testable	No event data	-	
DQ024	Task state transitions	Transformation	Not testable	No task attempt history	-	
DQ025	Reward state transitions	Transformation	Not testable	No reward data	-	
DQ026	Withdrawal state transitions	Transformation	Not testable	No withdrawal data	-	
DQ027	Referral stage order	Transformation	Not testable	No referral data	-	
DQ028	Reversal link valid	Application	Not testable	No reversal data	-	
DQ029	Self-referral	Application	Not testable	No referral data	-	Self-referral cannot be checked without a referrer id
DQ030	Reward-wallet agreement	Analytics Layer	Not testable	No reward or wallet data	-	
DQ031	Wallet balance agreement	Analytics Layer	Not testable	No wallet data	-	
DQ032	Withdrawal-wallet agreement	Analytics Layer	Not testable	No withdrawal data	-	
DQ033	Exchange pairing	Transformation	Not testable	No wallet data	-	
DQ034	Task-reward link agreement	Analytics Layer	Not testable	No task or reward data	-	

Rule	Name	Primary layer	Result	Detail	Issue ID(s)	Note
DQ035	No mixed-unit totals	Dashboard	Not testable	No dashboards	-	A design rule, not a data rule
DQ036	Data freshness	Raw Data	Not testable	No receive times	-	The file is a one-off extract
DQ037	Late-arriving data	Data Collection	Not testable	No event or receive times	-	
DQ038	Volume anomaly	Raw Data	Information	Audit sheet has 1,008 rows against 1,000 distinct users (8 extra rows)	-	Matches the 8 repeated rows
DQ039	Simulated and production data separated	Raw Data	Partly	Every audit row is labelled 'SIMULATED - NOT ACTUAL VELOOP DATA'; the clean sheet has no per-row label	-	Recommend the same label on the clean sheet

5. Rules proposed by this audit

Rule	Name	Issue type	Applies to	Logic	Layer	Backstop	Action	Severity	Owner	Evidence
DQ040	Derived-field consistency	inconsistent derived value	Any stored derived field (for example days_since_registration)	Recompute the derived value from its source fields and compare; a difference above a tolerance is flagged	Transformation	Analytics Layer	Flag and load	High	Data Analyst	Found by DQA-12
DQ041	Counter consistency	impossible state	lifetime_sessions, active_days, last_login_datetime, xp_balance	Sessions above 0 require a login time; active_days cannot exceed the days available or be 0 when sessions exist; XP above 0 requires activity	Transformation	Analytics Layer	Flag and load	Medium	Data Analyst	Found by several DQA issues
DQ042	Consent consistency	impossible state	consent_status, marketing_opt_in	Marketing opt-in requires consent_status = Accepted	Application	Transformation	Reject at source	High	Product / Legal	Found by DQA-19
DQ013x	Extend DQ013 to non-negative numeric fields	negative values where inappropriate	Derived and counter fields	Value 0 or more where the field is a count or a duration	Application	Transformation	Reject at source	Medium	Engineering	Found by DQA-13
DQ039x	Per-row simulation label	environment separation	Every simulated dataset	Each row carries a data_environment label, including the clean sheet	Raw Data	Analytics Layer	Block	Critical	Data Analyst	Partly met: only the audit sheet has the label

6. Recommendations

#	Recommendation	Evidence	Owner	Layer
1	Make user_id unique in the store, not just in the file (DQ002), and write idempotently (DQ018).	DQA-01	Engineering	Application
2	Generate registration and login times on the server in UTC with an explicit offset; reject non-ISO values (DQ004, DQ005, DQ006).	DQA timestamps	Engineering	Application / Collection
3	Do not store derived values (days_since_registration, is_active_7d, is_active_30d). Compute them in the analytics layer from events, or recompute and compare on load (DQ040).	DQA derived value; negative days	Data Analyst	Transformation
4	Compute counters (sessions, active days, XP) from events, and check them against each other (DQ041).	DQA counter issues	Data Analyst	Transformation
5	Enforce 'marketing opt-in requires accepted consent' at the application (DQ042) and confirm the rule with Legal.	DQA consent	Product / Legal	Application
6	Keep account state (stored) separate from activity (derived) and define 'Inactive'.	DQA status vs activity	Product	Design
7	Add a referrer link (referral_code_used and a Referrals table) so referral conversion can be measured.	Profiling F07	Engineering	Design
8	Label every simulated row with data_environment, including the clean sheet (DQ039).	Rule coverage	Data Analyst	Raw Data
9	Document every column in the dictionary (platform and country are missing) and state the reference time of derived fields.	DQA undocumented fields	Data Analyst	Documentation
10	Minimise personal fields: share only sanitised data with analysts and interns.	DQA personal fields	Product	Governance
11	Supply the remaining datasets so the 23 rules that could not be tested can run (A046).	Rule coverage	Mentor / Product	Data supply

7. Limitations

- Only the Users entity was supplied, so 23 of 39 rules could not be tested and no relationship beyond Users could be checked.
- Allowed value lists were taken from the clean sheet because no approved lists exist yet (A022).
- Likely causes for injected problems follow the file's README; causes for inherent issues are inferences, not facts.
- The rule about active_days assumes last_login_datetime is the most recent session.

Next: add DQ040 to DQ042 to the Data Quality Control Framework through the change log, and re-run this audit when the remaining datasets are supplied.